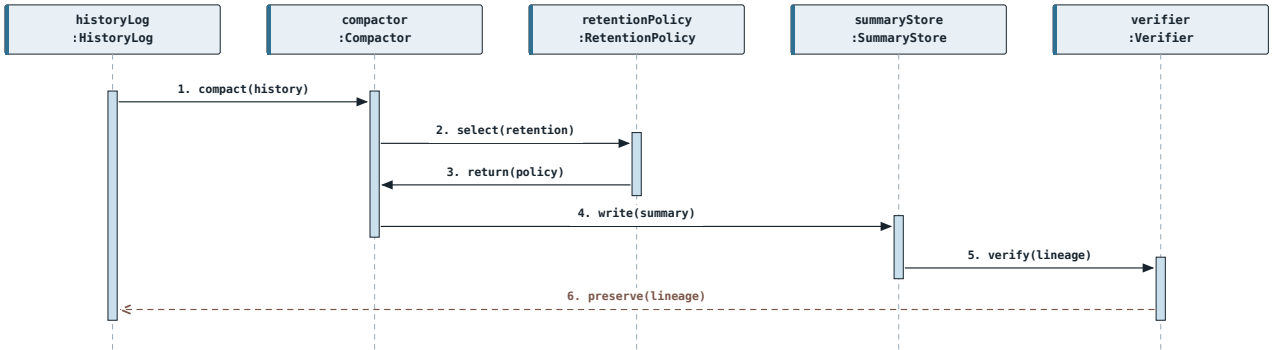
P27 Lineage and threshold compaction

Primitive / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Bound history while retaining useful state.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	History approaches a context or count threshold before a model call.
TRIGGER	A compaction policy activates.
STRUCTURE	Threshold, selection strategy, summarizer or truncator, and lineage metadata.
EFFECT	Model input is reduced while retained and removed evidence remains attributable.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Context usage, protected items, and eviction policy are available.
ENSURES	The replacement fits the target and records what replaced which history.
INVARIANT	Required system, safety, or current-task evidence is not silently evicted.
VIOLATION	Summary failure, threshold conflict, or empty projection falls back explicitly.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.